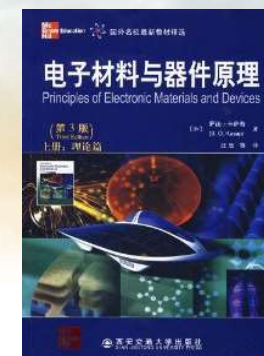
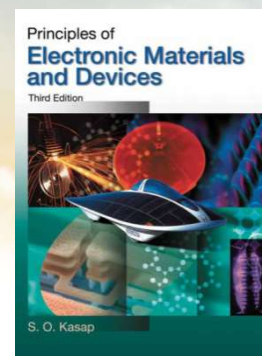




南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》 Solid-State Electronics



Ch4-1 Modern Theory of Solids

Dr. Rui CHEN (陈锐)

Email: chenr@sustech.edu.cn

Office: Room 428 at College of
Engineering South Tower

Tel: 0755-88018563



Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

Previously on Quantum Mechanics

- Using Schrödinger's wave equation to describe the behavior of single electron

$$\frac{d^2\varphi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V)\varphi(x) = 0$$

- Probability density distribution function $|\varphi(x, y, z)|^2$
- In atomic level, the electron's energy and momentum are quantized

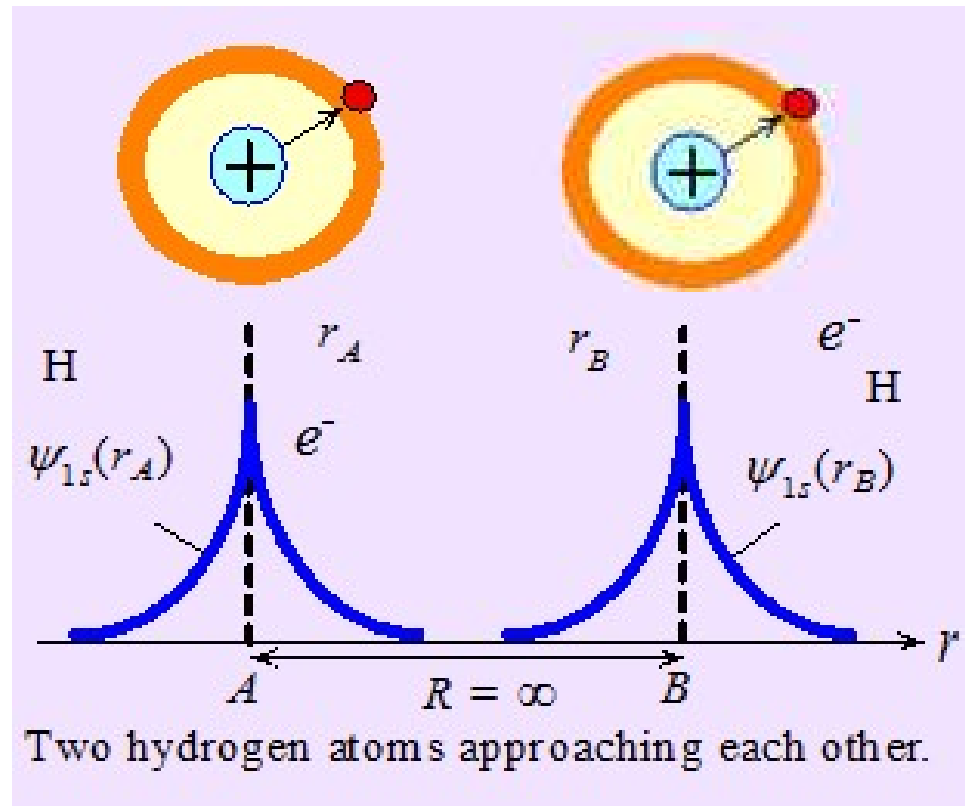
$$E_n = -\frac{me^4Z^2}{8\varepsilon_0^2\hbar^2n^2}$$

- 1) Principal quantum number $n = 1, 2, 3, \dots$
- 2) Orbital angular momentum quantum number $l = 0, 1, 2, \dots (n-1)$
- 3) Magnetic quantum number $m_l = -l, -(l-1), \dots, 0, \dots, (l-1), l$
- 4) Spin quantum number $S = \pm 1/2$

What is the situation in the case of solid? When many atoms combine together?



In the Case of H Atom



$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$$

$$\psi_{100}(r, \theta, \phi) = \frac{1}{2\sqrt{\pi}} \left(\frac{1}{a_0} \right)^{3/2} 2 \exp\left(-\frac{r}{2a_0} \right)$$

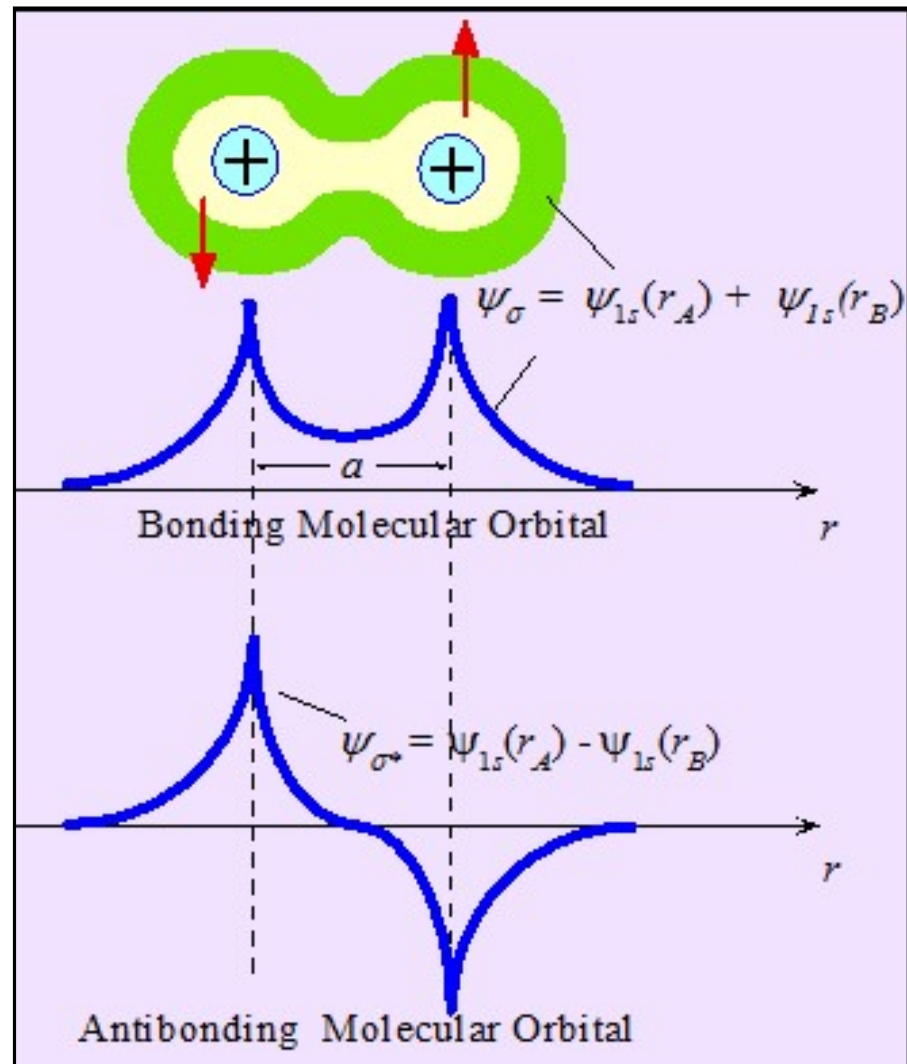
- 1) Can they have two sets of identical atomic orbitals?
- 2) Will remain the same as H?

As they approach:

- Electron interact with each other and with the other nuclei, potential energy change
- The electrons should obey a new wave function



In the Case of H₂



As they approach:

- Potential energy change
- Wave function overlaps: as a result, they interfere with each other constructive (in phase) or destructive (out of phase).

$$\psi_{\sigma} = \psi_{1s}(r_A) + \psi_{1s}(r_B) \quad \text{In phase, bonding}$$

$$\psi_{\sigma^*} = \psi_{1s}(r_A) - \psi_{1s}(r_B) \quad \text{Out of phase, anti-bonding}$$

The method we used for generating two separate molecular orbitals is called the **Linear Combination of atomic Orbitals (LCAO)** method, since it came from the linear combination of two identical atomic orbitals.

原子轨道线性组合

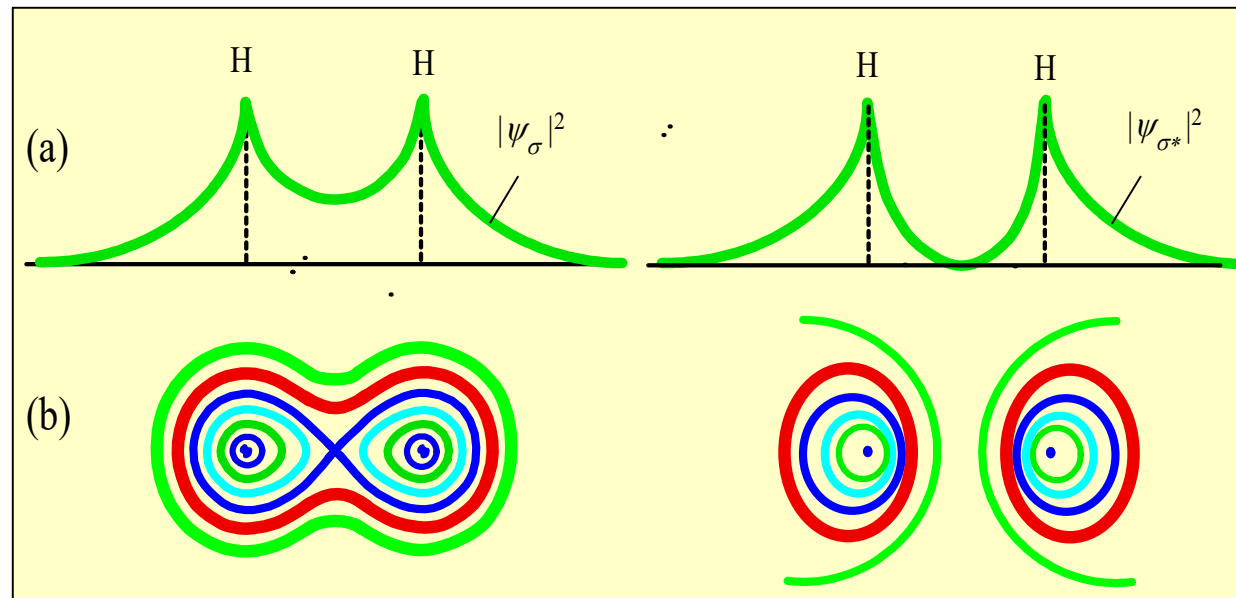


南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

In the Case of H₂

- Attraction: electron and proton
- Repulsion: proton-proton, electron-electron
- Bonding: energy of attraction is stronger than repulsion
- Anti bonding: energy of repulsion is stronger than attraction

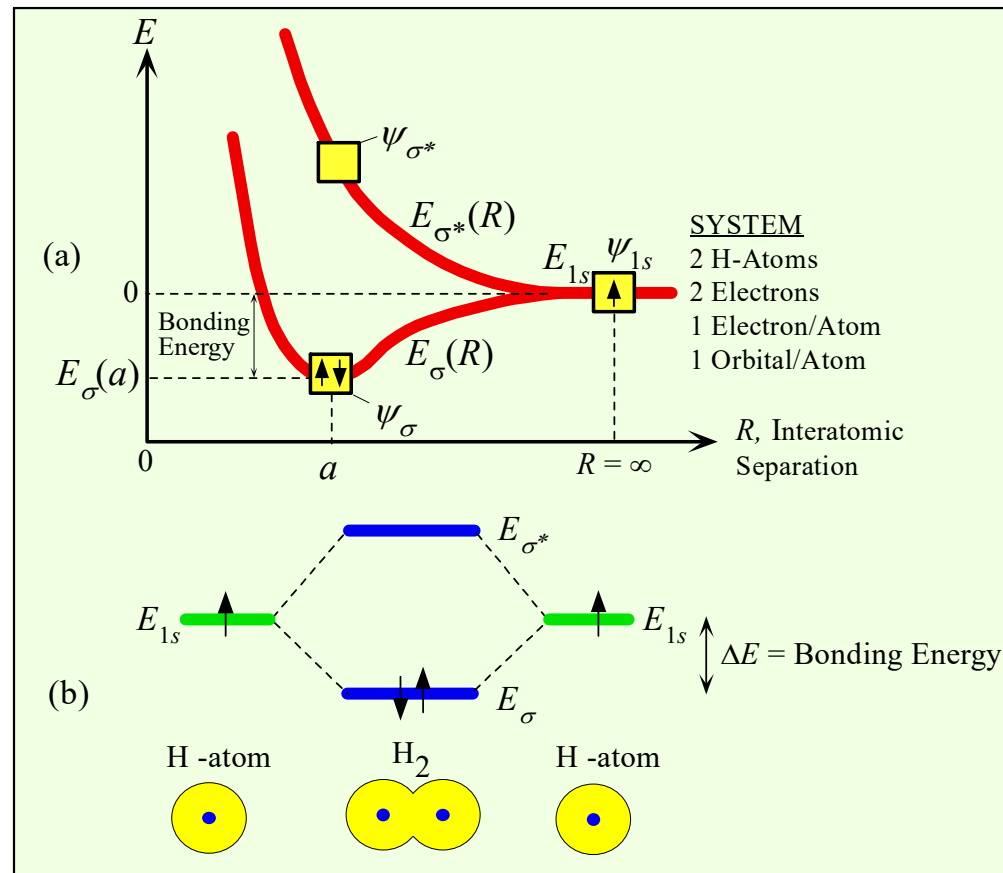
How about the energy of each orbital?



The splitting of a one-atom energy level when a molecule is formed is analogous to the splitting of the resonant frequency in an RLC circuit when two such circuits are brought together and coupled.



In the Case of H₂



- The energy difference between the E of the two isolated H atoms and the E_{σ} minimum energy at $R=a$ is the **bonding energy**
- The wavefunction φ_{σ} corresponding to the lowest electron energy is called the **bonding orbital**, and φ_{σ^*} is the **antibonding orbital**
- When **two atoms** are brought together, the two identical atomic wavefunction combine in two ways to generate **two different molecular orbitals**, each with a **different energy**

H-atom: attractive energy

H₂ molecule: attractive energy (-) and repulsive energy (+), total energy is lower than 2 H-atoms

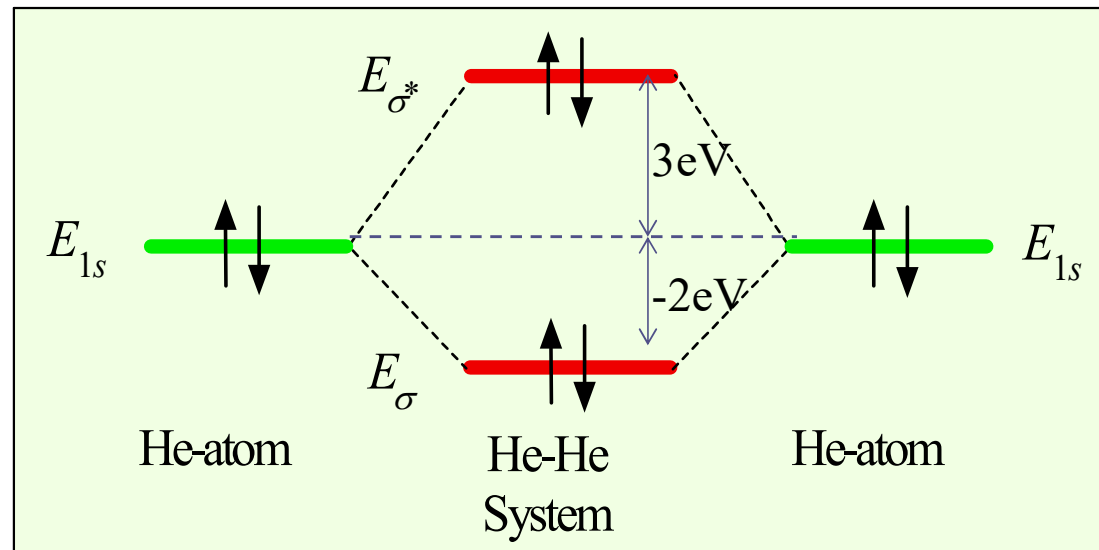
H₂⁻ is possible

How about H₂²⁻



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

In the Case of He



He₂ and He, which one is more stable? Why?

Answer: 2 He atoms have 4 electrons. When He atoms come together, 2 of the electrons enter the E_{σ} and 2 the E_{σ}^* levels so that the overall energy is greater than two isolated He atoms.

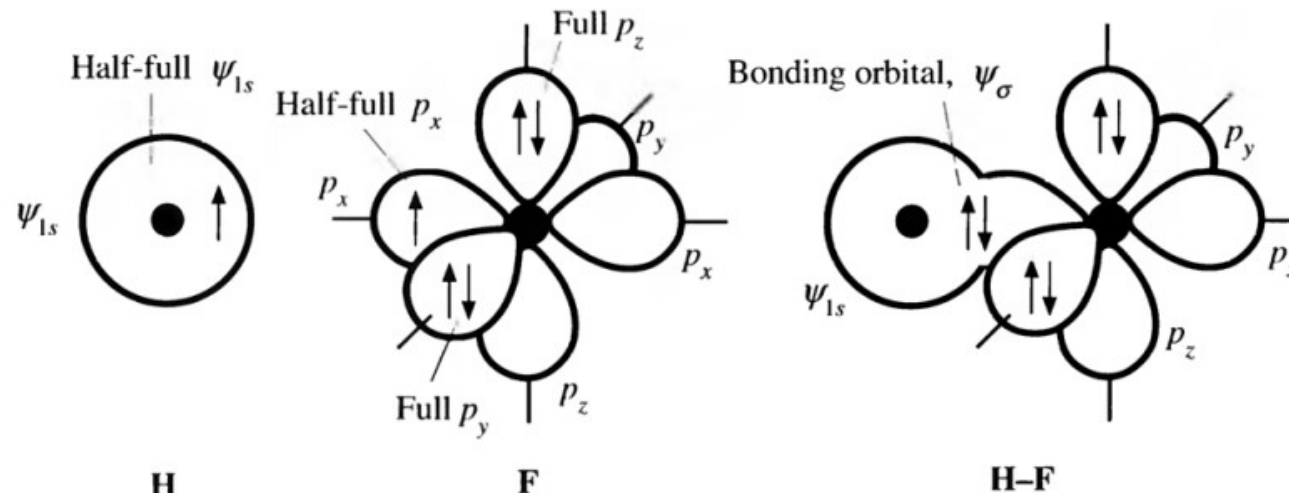
- In fact, **full orbitals** repel each other, because an overlap results in an increase in the system energy
- To form a bond between two atoms, need an overlap of half-occupied orbitals.



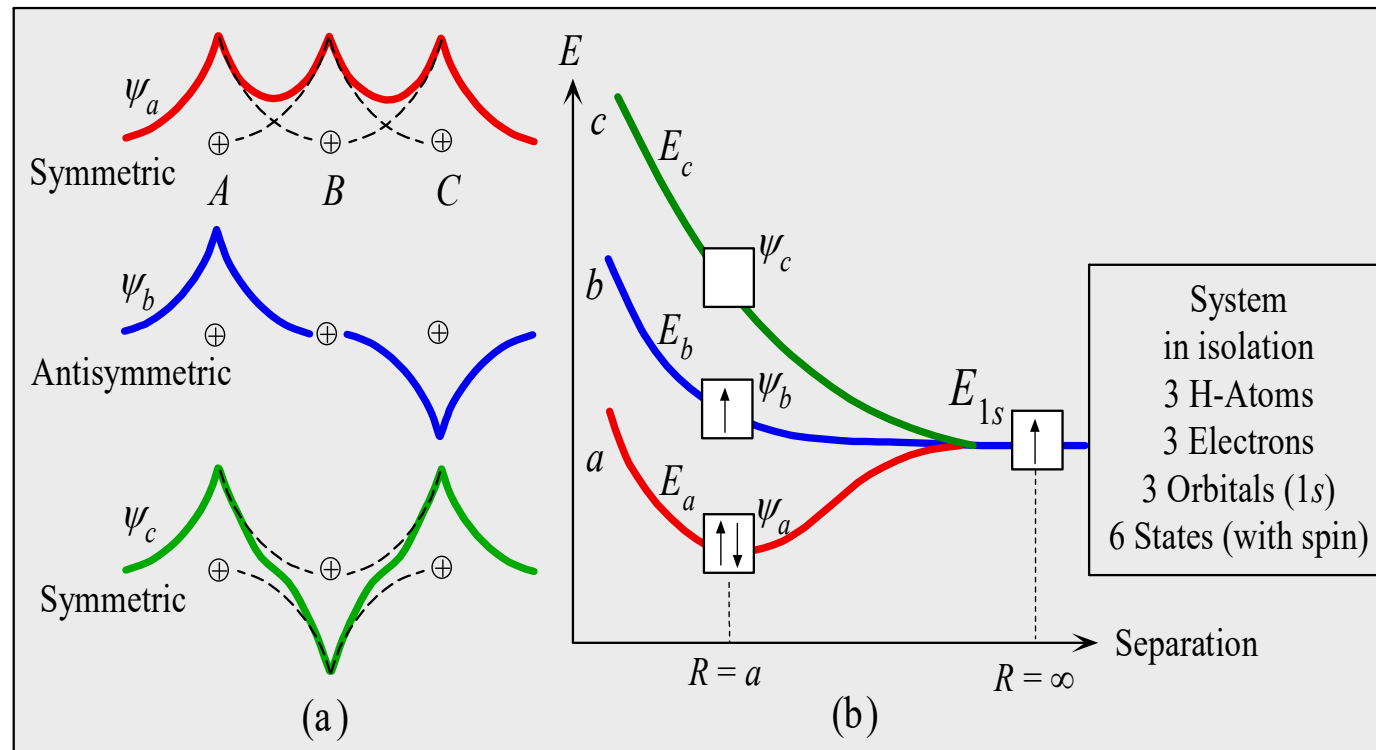
In the Case of HF

EXAMPLE 4.1

HYDROGEN HALIDE MOLECULE (HF) We already know that H has a half-occupied $1s$ orbital, which can take part in bonding. Since the F atom has the electronic structure $1s^2 2s^2 p^5$, two of the p orbitals are full and one p orbital, p_x , is half full. This means that only the p_x orbital can participate in bonding. Figure 4.6 shows the electron orbitals in both H and F. When the H atom and the F atom approach each other to form an HF molecule, the ψ_{1s} orbital of H overlaps the p_x orbital of F. There are two possibilities for the overlap. First, ψ_{1s} and p_x can overlap in phase (both positive or both negative), to give a ψ_σ orbital that does not have a node between H and F, as shown in Figure 4.6. Second, they can overlap out of phase (one positive and the other negative), so that the overlap orbital ψ_{σ^*} has a node (similar to ψ_{σ^*} in Figure 4.1). We know from hydrogen atomic wavefunctions in Chapter 3 that orbitals with more nodes have higher energies. The molecular orbital ψ_σ therefore corresponds to a bonding orbital with a lower energy than the ψ_{σ^*} orbital. The two electrons, one from ψ_{1s} and the other from p_x , enter the ψ_σ orbital with spins paired, thereby forming a bond between H and F.



In the Case of H₃



Q1:
Each molecular orbit must be symmetrical or antisymmetrical why?

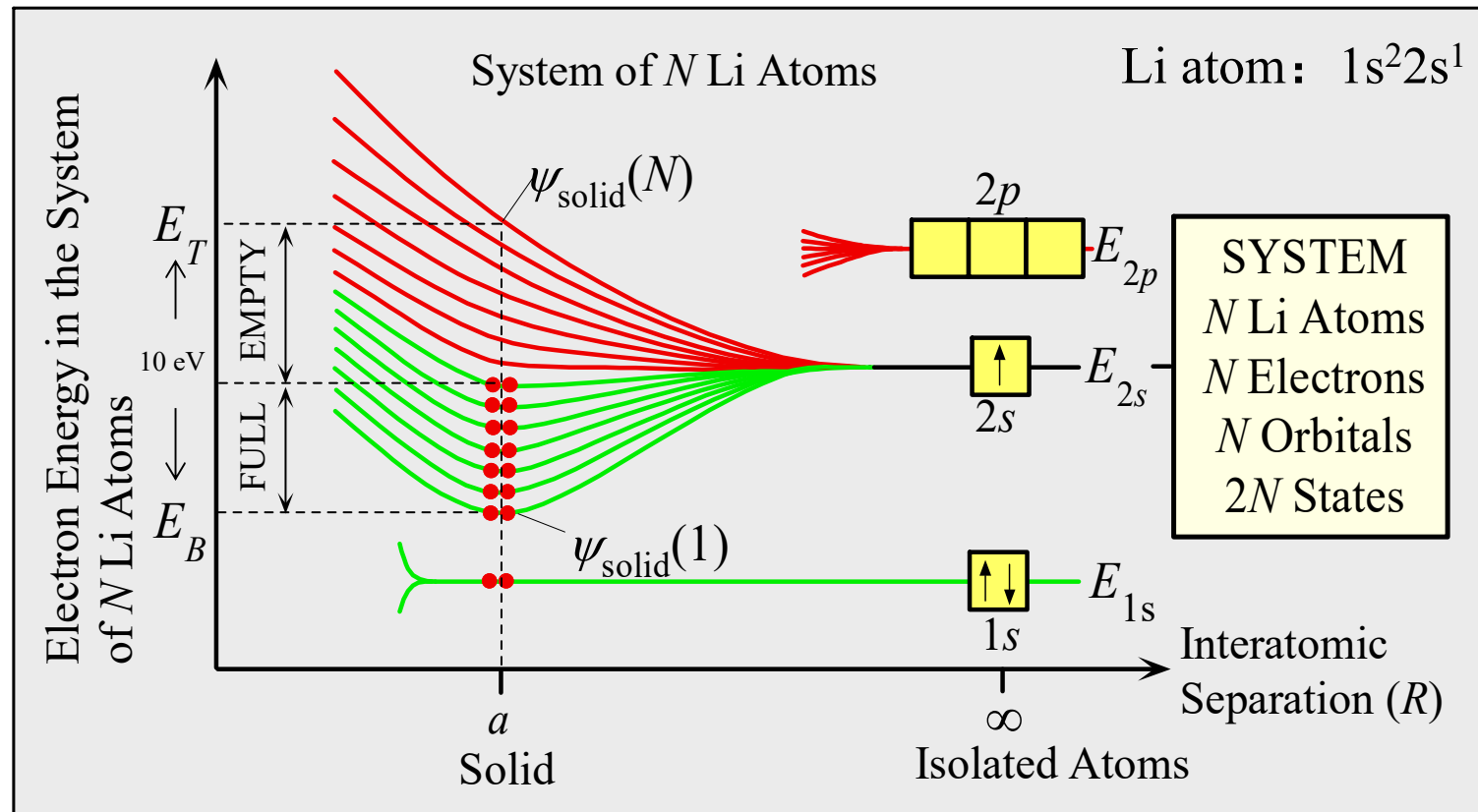
Q2:
H₃ and H₂ which one has higher energy?

- When three H atoms are brought together, the potential energy for A, B, C atom is different, depending on their position.
- Resulting in three different wave functions
- Electrons sit the orbital with lower energy first

$$\begin{aligned}\psi_a &= \psi_{1s}(A) + \psi_{1s}(B) + \psi_{1s}(C) \\ \psi_b &= \psi_{1s}(A) - \psi_{1s}(C) \\ \psi_c &= \psi_{1s}(A) - \psi_{1s}(B) + \psi_{1s}(C)\end{aligned}$$



In the Case of Solid



- As N Li atoms approach together, each $2s$ electron has different potential energy, depending on their position.
- Resulting in N different wave function. Hence the energy level of $2s$ splits into N finely separated energy levels.
- There are N $2s$ electrons, but $2N$ states. Electrons sit the orbital with lower energy first.



Formation of Energy Band

N atoms brought together

→ each level → N discrete

closely spaced energy levels

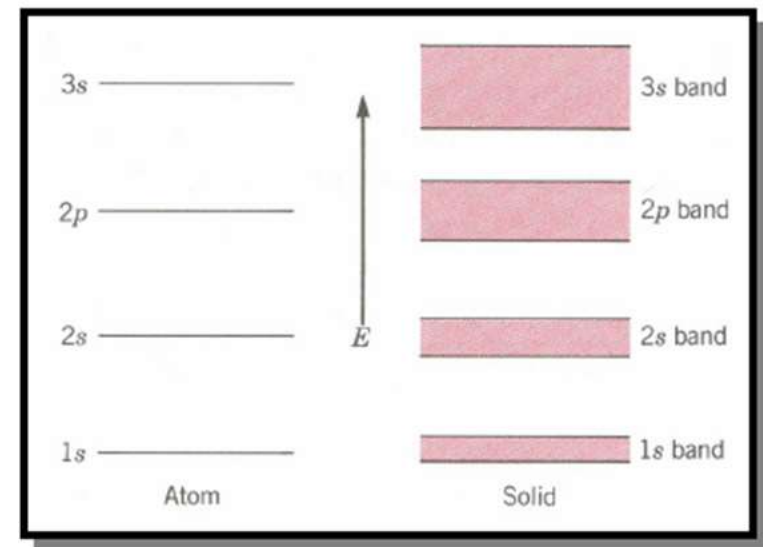
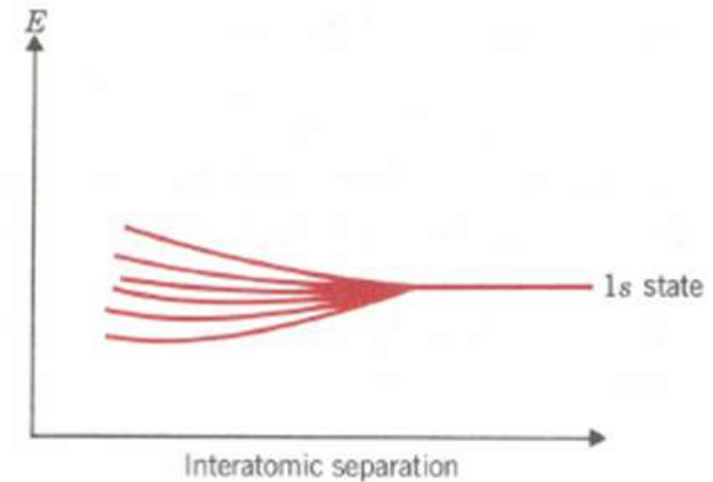
→ a quasi-continuous band of energy levels

For example,

width of a band, ~ a few eV

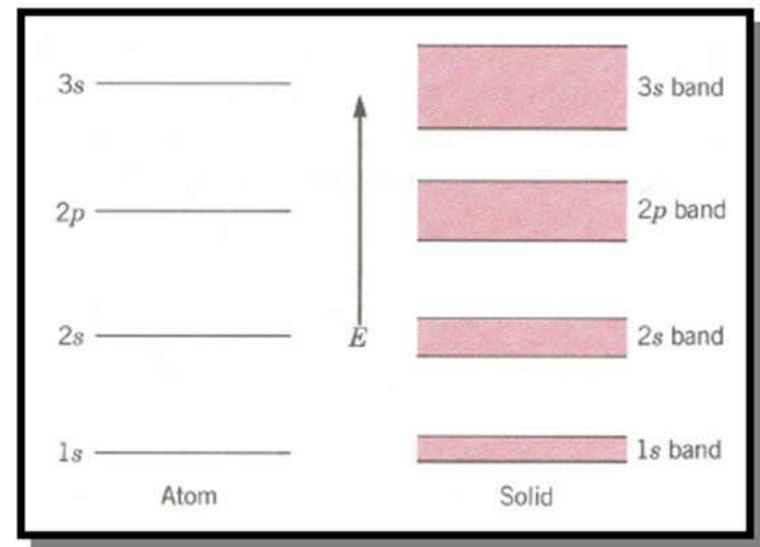
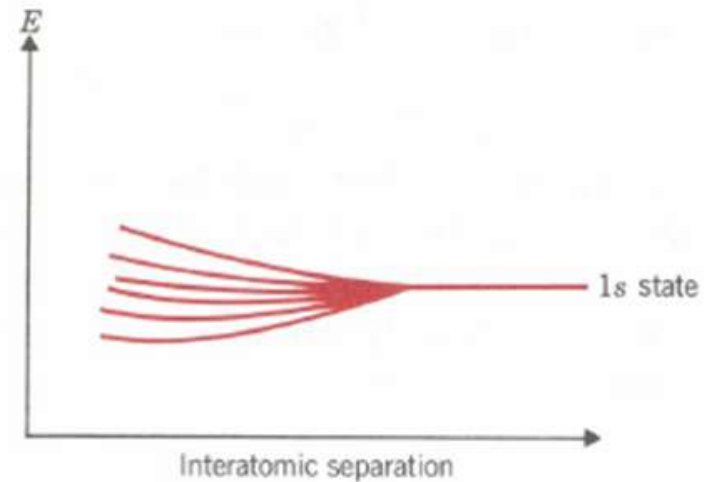
If $N=10^{23}$

→ Separation between adjacent level $\sim 10^{-23}$ eV

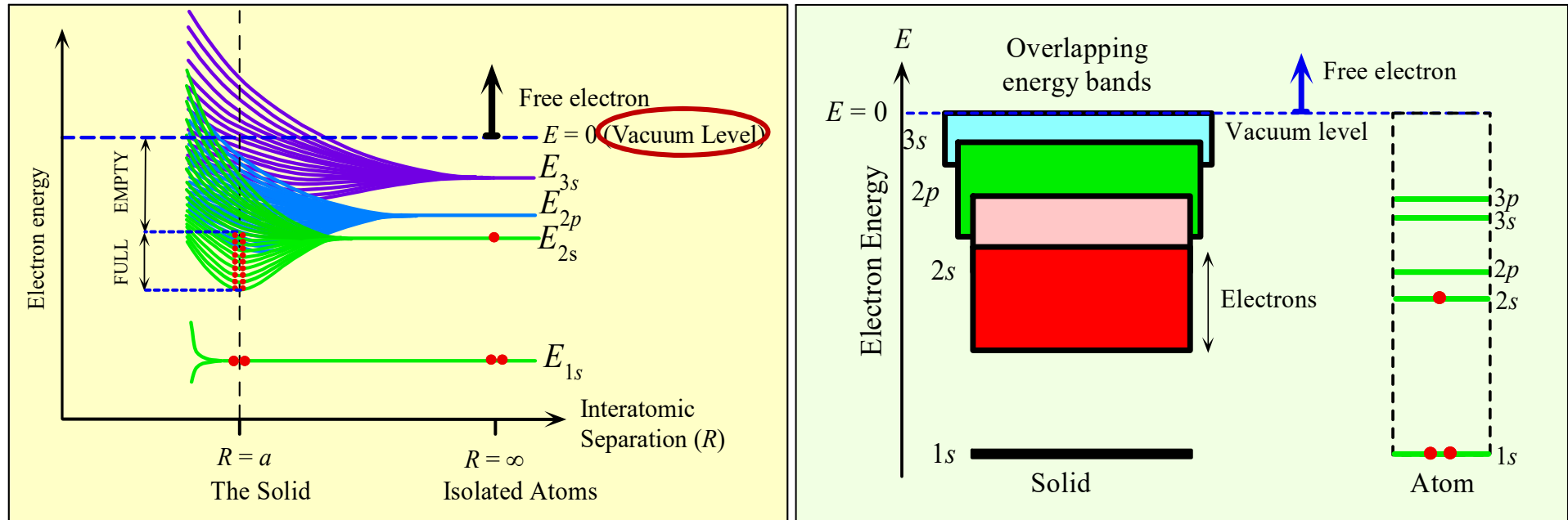


Formation of Energy Band

- Width of the bands should not depend appreciably on N
- Widths of the bands depends mainly on distance between adjacent atoms
 - ➔ Atoms closer to each other
 - ➔ greater bandwidth
- Bandwidth of the low-lying levels < bandwidth of the higher energy levels
- Interaction stronger ➔ larger bandwidth



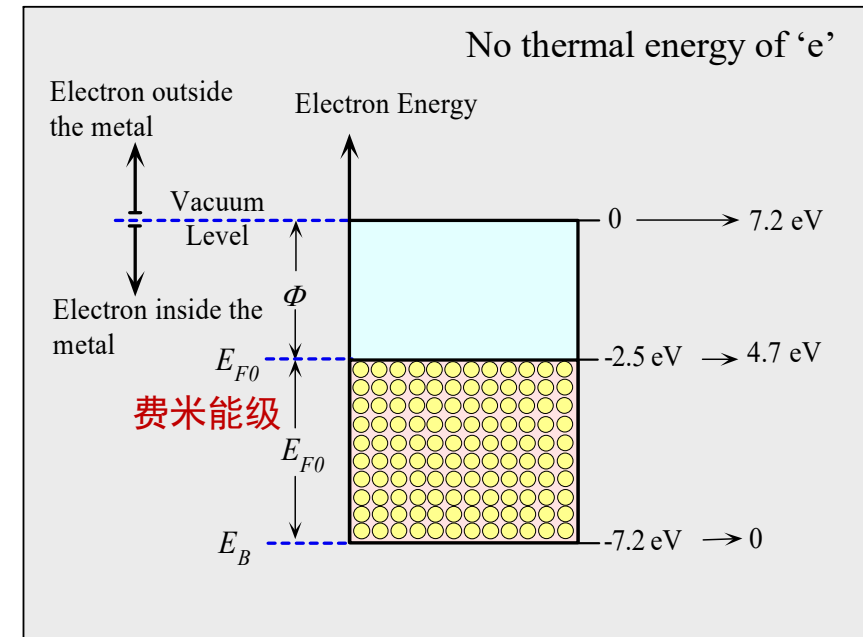
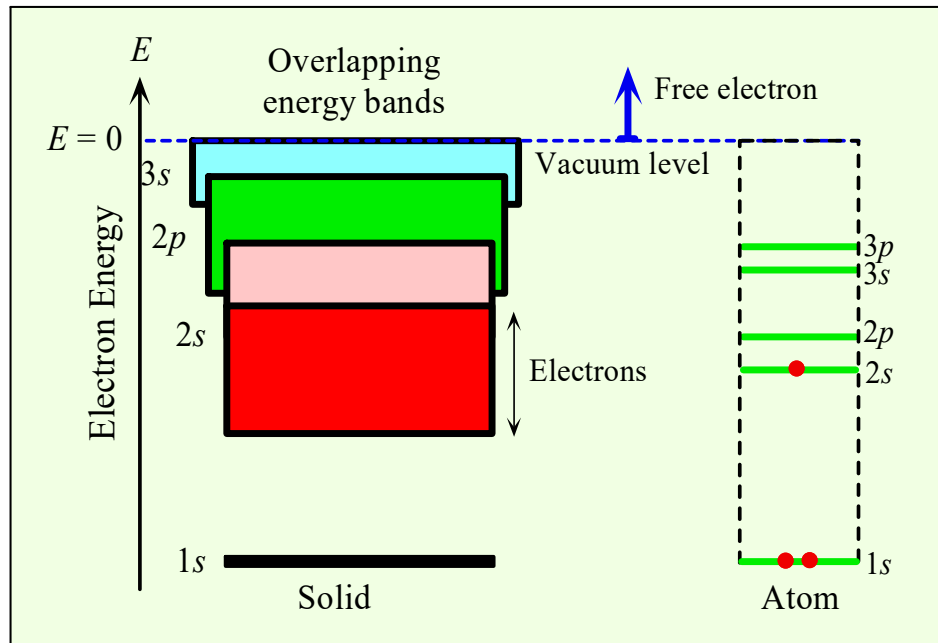
Metal: Bands Overlap



- $2p$, $3s$ and other higher energy level, also split into finely separated energy levels.
- They overlap with $2s$ band, hence provide further energy levels and extend the $2s$ band into higher energy levels.



Fermi Level and Work Function



- At absolute zero, all the energy levels up to the **Fermi level** are full.
- The energy required to excite an electron from the Fermi level to the vacuum level is called work function.

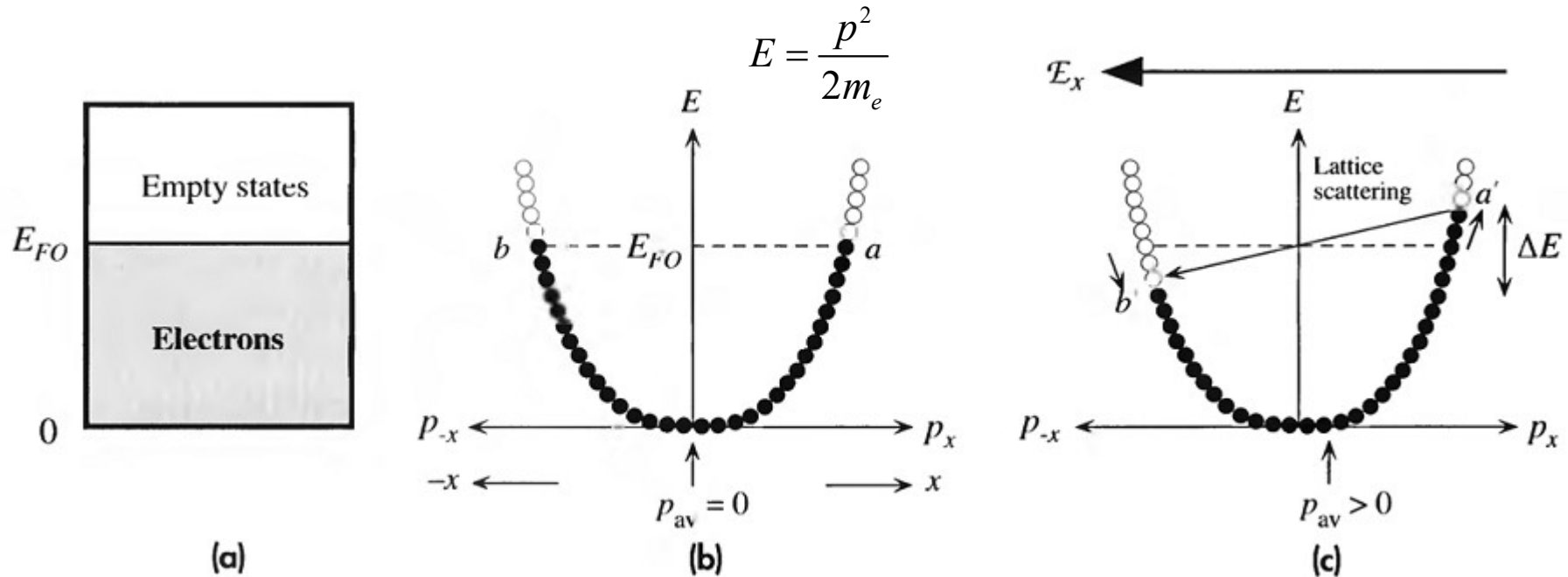
Table 4.1 Fermi energy and work function of selected metals

	Metal							
	Ag	Al	Au	Cs	Cu	Li	Mg	Na
Φ (eV)	4.5	4.28	5.0	2.14	4.65	2.3	3.7	2.75
E_{F0} (eV)	5.5	11.7	5.5	1.58	7.0	4.7	7.1	3.2



Conduction in Metal

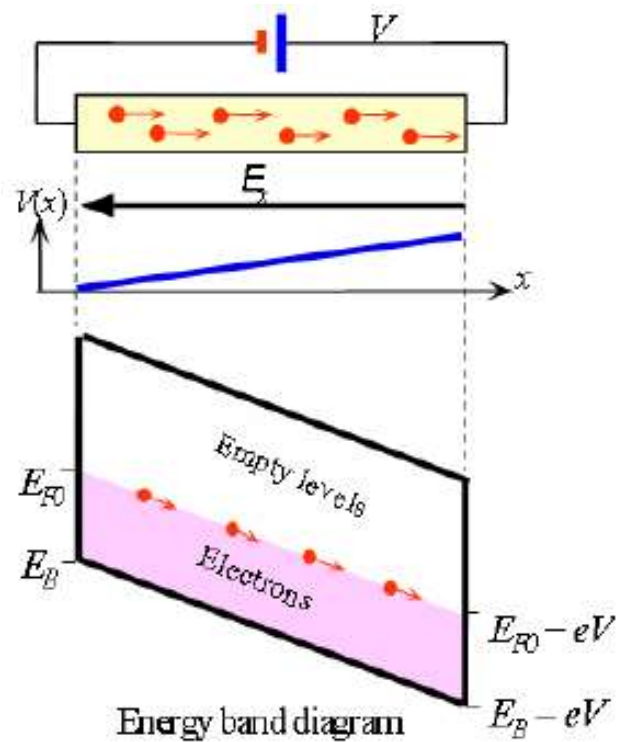
能带中电子的性质



- Energy band diagram of a metal
- In the absence of a field, there are as many electrons moving right as there are moving left. The motions of two electrons at each energy cancel each other as for a and b
- In the presence of a field in the $-x$ direction, the electron a accelerates and gains energy to a' where it is scattered to an empty state near E_{F0} but moving in the $-x$ direction. The average of all momenta values is along the $+x$ direction and results in a net electric current



Conduction in Metal



Conduction in a metal is obtained by the drift of electrons around the Fermi Level in the presence of E . When a voltage is applied, the energy band is bent to be lower at the positive terminal so that the electrons' potential energy decreases as it moves toward the positive terminal.

Notice that the application of the electric field bends the energy band, because electrostatic PE of the electron is $-eV(x)$ where $V(x)$ is the voltage at position x .

Addition of heat to a metal

- can excite electrons in the band to higher energy levels and can be absorbed by the conduction electrons of a metal
- can increase the amplitude of atomic vibrations

Which contributes to the heat capacity of a metal? Which one is dominated?

- Lattice vibration: room temperature
- Conduction electron: lowest temperatures



Energy Band of Metal: Na

For example:

Na crystal: $1s^2 2s^2 2p^6 3s^1$ (11 electrons/atom)

N atoms is in the solid

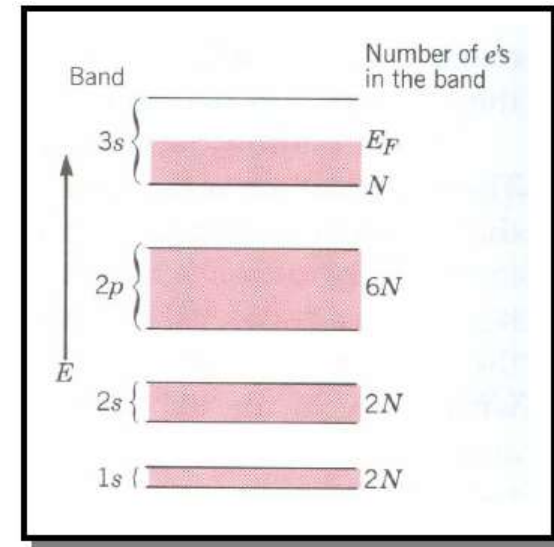
→ Total of $11N$ electrons

When electric field applied

→ electron gain energy

→ move into empty, slightly higher energy states

→ current conduction



Energy Band of Metal: Mg

Magnesium has two 3s electrons.

Mg crystal: $1s^2 2s^2 2p^6 3s^2$ (12 electrons/atom)

You expect the 3s band to be full, the 3p band to be empty, with a forbidden gap in between.



Magnesium is conductor or insulator? Why?



Energy Band of Metal: Mg

But magnesium is a metal (actually, a “semimetal”)

The 3p and 3s bands overlap. There are many empty states nearby into which electrons can be accelerated.



For example:

Mg crystal: $1s^2 2s^2 2p^6 3s^2$ (12 electrons/atom)

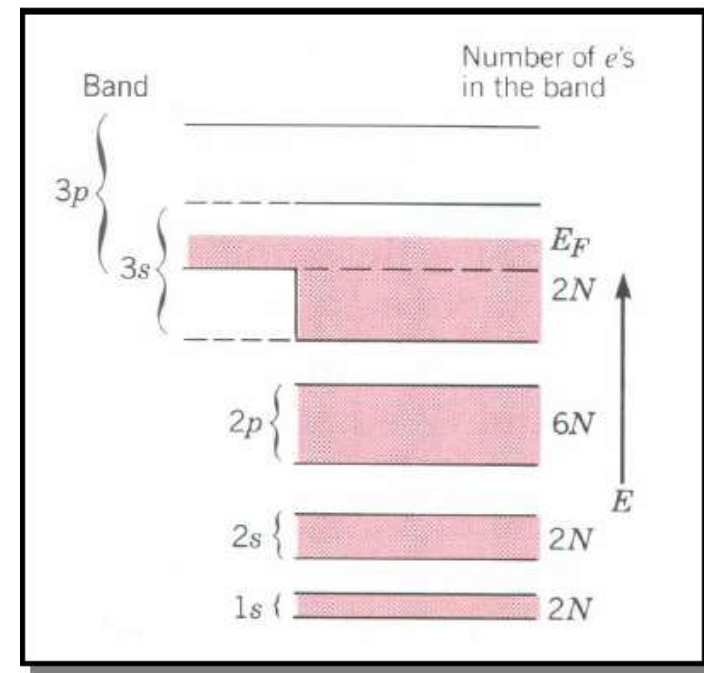
N atoms is in the solid

→ Total of $12N$ electrons

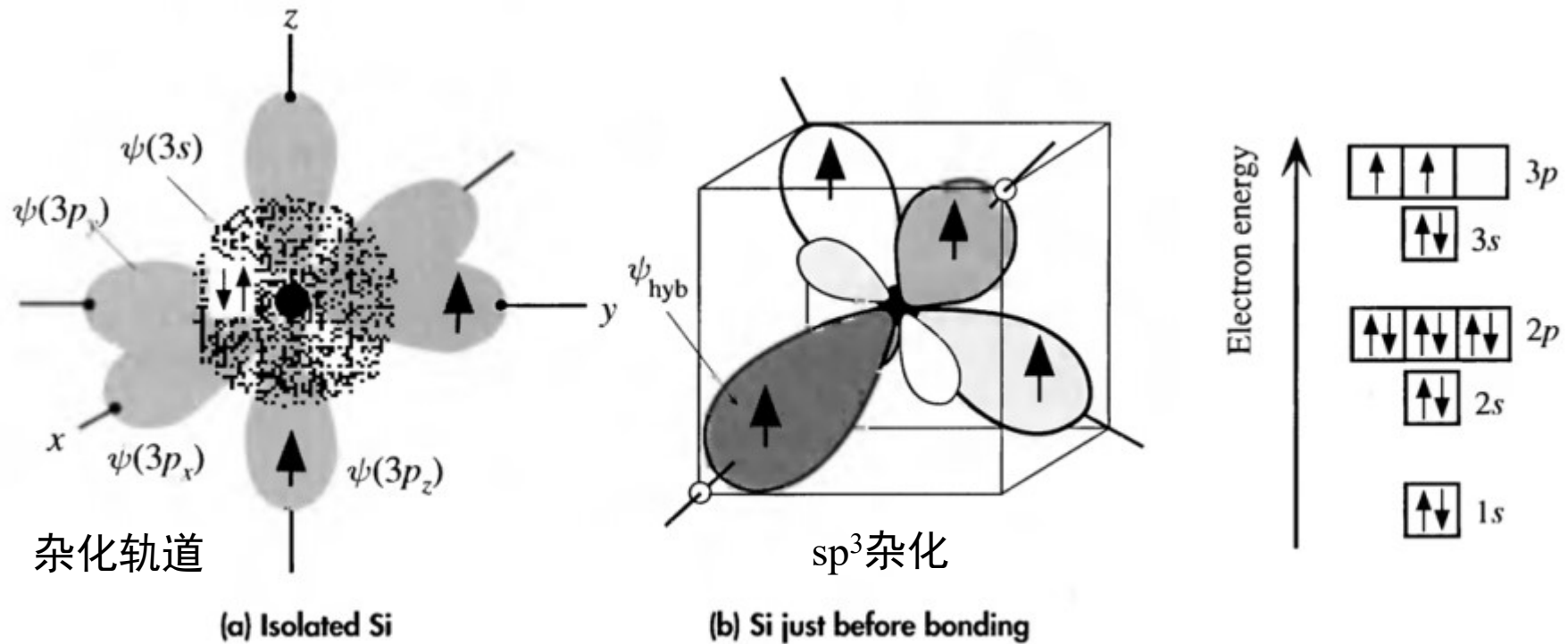
For Mg, 3p band and 3s band overlap

When electric field applied

- electron gain energy
- move into empty, slightly higher energy states
- current conduction
- Mg crystal is a conductor



4.3 Semiconductor: sp^3 Hybridization



- Si is in Group IV in the Periodic Table. An isolated Si atom has two electrons in the 3s and two electrons in the 3p orbitals.
- When Si is about to bond, the one 3s orbital and the three 3p orbitals become perturbed and mixed to form four **hybridized orbitals**, ϕ_{hyb} , called sp^3 orbitals, which are directed toward the corners of a tetrahedron. The ϕ_{hyb} orbital has a large major lobe and a small back lobe. Each ϕ_{hyb} orbital takes one of the four valence electrons.



Conduction Band and Valence Band

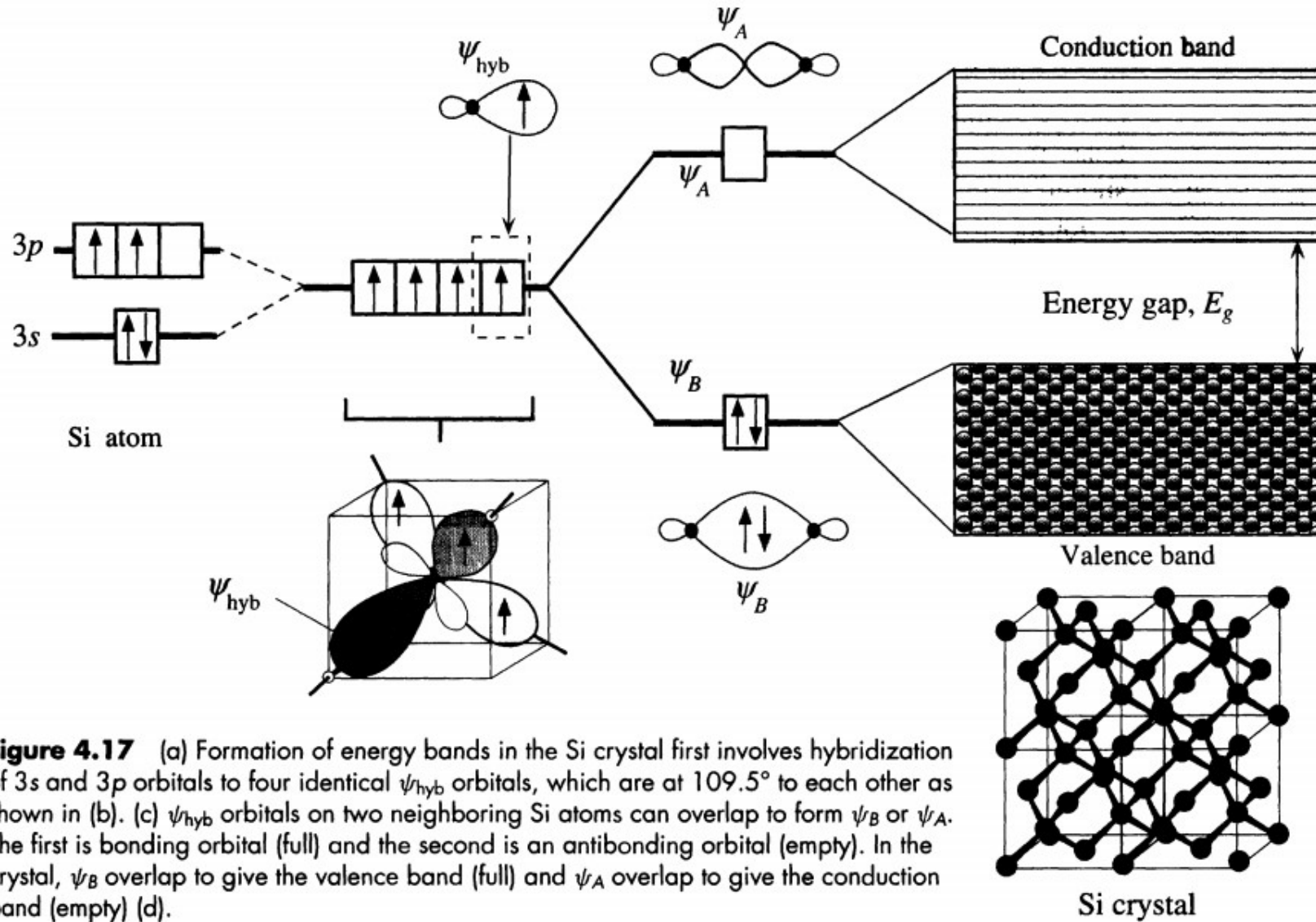


Figure 4.17 (a) Formation of energy bands in the Si crystal first involves hybridization of 3s and 3p orbitals to four identical ψ_{hyb} orbitals, which are at 109.5° to each other as shown in (b). (c) ψ_{hyb} orbitals on two neighboring Si atoms can overlap to form ψ_B or ψ_A . The first is bonding orbital (full) and the second is an antibonding orbital (empty). In the crystal, ψ_B overlap to give the valence band (full) and ψ_A overlap to give the conduction band (empty) (d).



Conduction Band and Valence Band

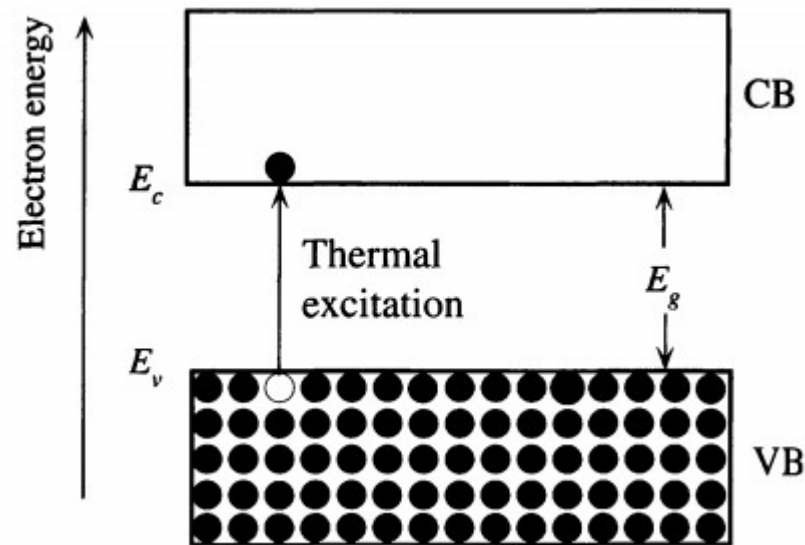


Figure 4.18 Energy band diagram of a semiconductor

CB is the conduction band and VB is the valence band. At 0 K, the VB is full with all the valence electrons.

CUTOFF WAVELENGTH OF A Si PHOTODETECTOR What wavelengths of light can be absorbed by a Si photodetector given $E_g = 1.1$ eV? Can such a photodetector be used in fiber-optic communications at light wavelengths of $1.31 \mu\text{m}$ and $1.55 \mu\text{m}$?

EXAMPLE 4.5

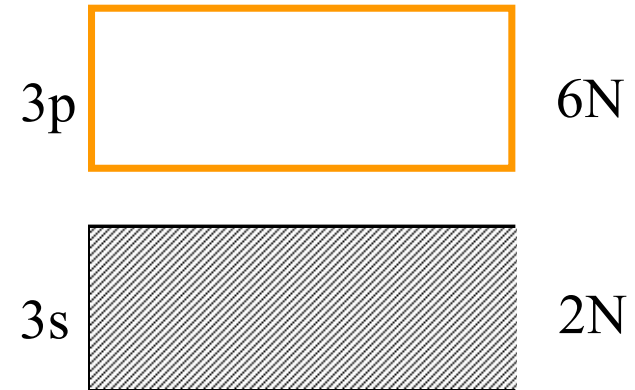
$$\lambda < \frac{hc}{E_g} = \frac{(6.6 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})}{(1.1 \text{ eV})(1.6 \times 10^{-19} \text{ J/eV})} \\ = 1.13 \times 10^{-6} \text{ m} \quad \text{or} \quad 1.1 \mu\text{m}$$

Since optical communications networks use wavelengths of 1.3 and $1.55 \mu\text{m}$, these light waves will not be absorbed by Si and thus cannot be detected by a Si photodetector.



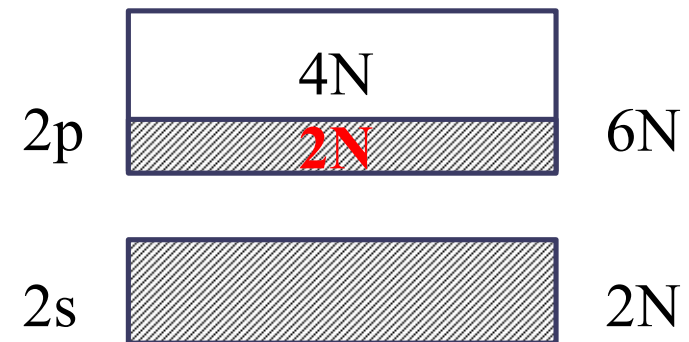
Energy Band of Insulator: Carbon

Materials which have bands either completely full or completely empty are insulators (unless band overlap occurs, as was the case for magnesium).



In a carbon atom, the 2p shell contains 2 electrons. There are 6 available states, so one would expect the 2p band to be 1/3 full* and carbon to be a conductor.

C crystal: $1s^2 2s^2 2p^2$
(6 electrons/atom)



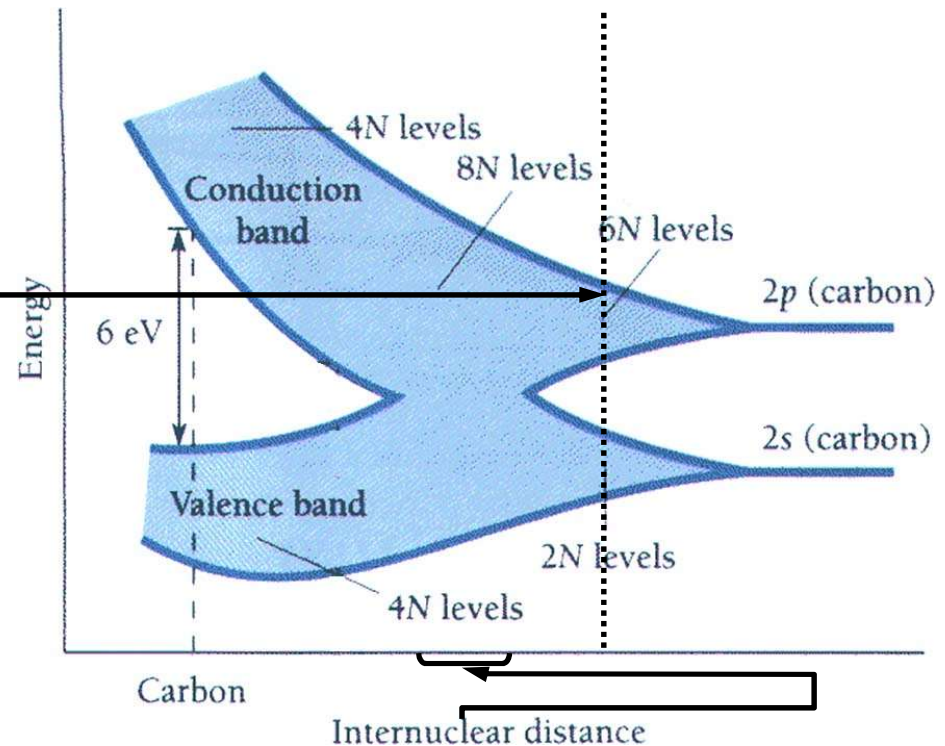
Carbon: Band Hybridization

- Figure shows energy bands in carbon (and silicon) as a function of interatomic separation.

At large separation, there is a filled 2s band and a 1/3 filled 2p band.

At this position, carbon is “metal”

C crystal: $1s^2 2s^2 2p^2$
(6 electrons/atom)



But electron energy can be lowered if the carbon-carbon separation is reduced.

There is a range of carbon-carbon separations for which the 2s and 2p bands overlap and form a hybrid band containing 8N states

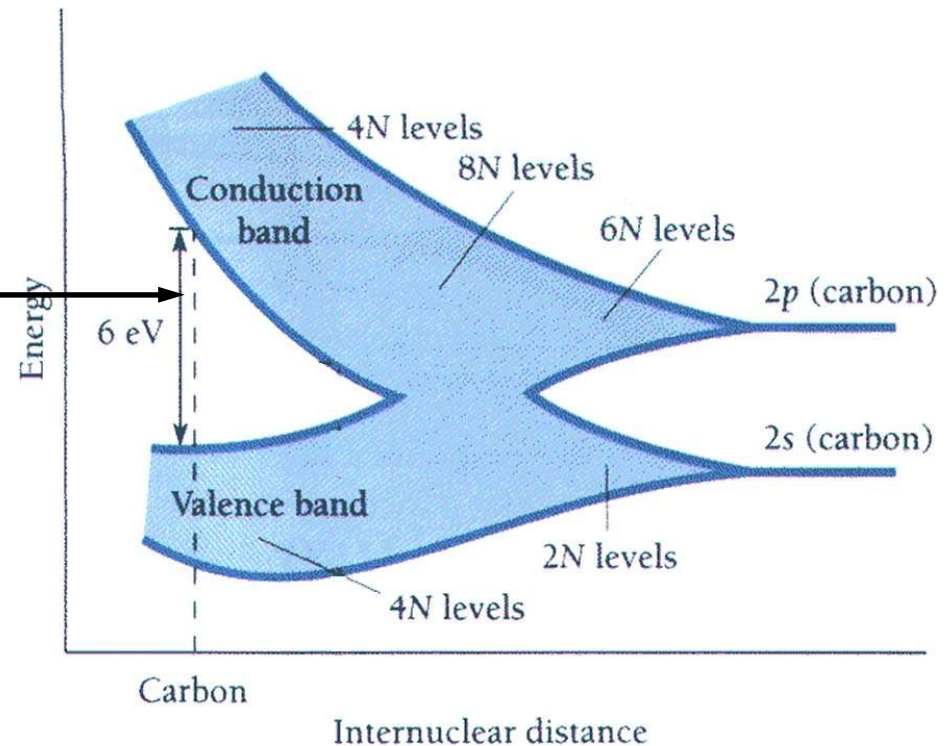


Carbon: Band Hybridization

But the minimum total electron energy occurs at this carbon separation.

At this separation there is a valence band containing $4N$ quantum states and occupied by $4N$ electrons.

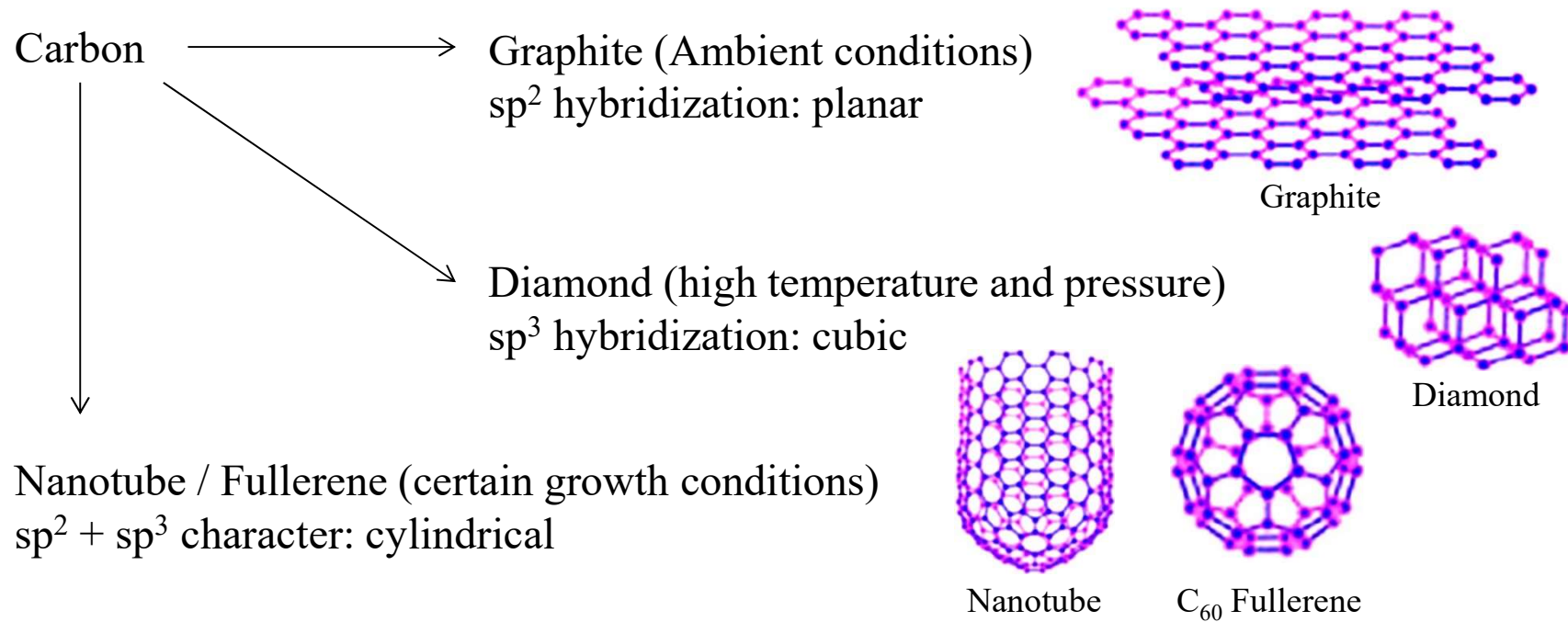
C crystal: $1s^2 2s^2 2p^2$
(6 electrons/atom)



The filled band is separated by a large gap from the empty conduction band. The gap is 6 eV — remember, $k_B T$ is about 0.025 eV at room temperature. The gap is too large for ordinary electric fields to move an electron into the conduction band. Carbon is an insulator in this area.



Why do Carbon Nanotubes form?

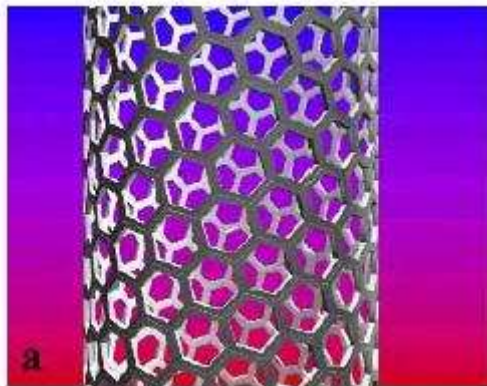
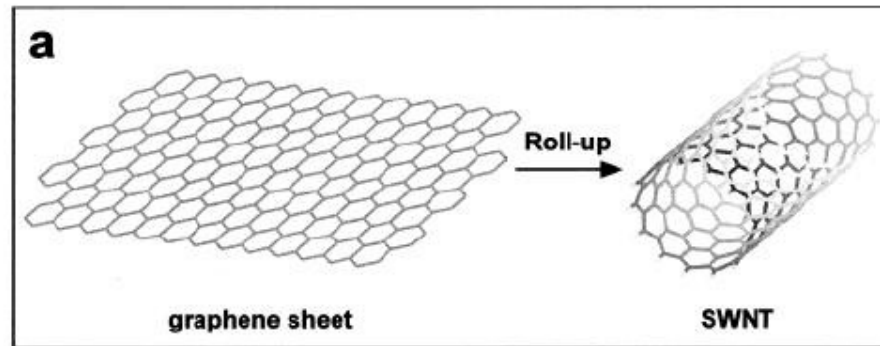


Nanotube formation { Eliminates dangling bonds
+
Increases strain energy } → Total energy decreases

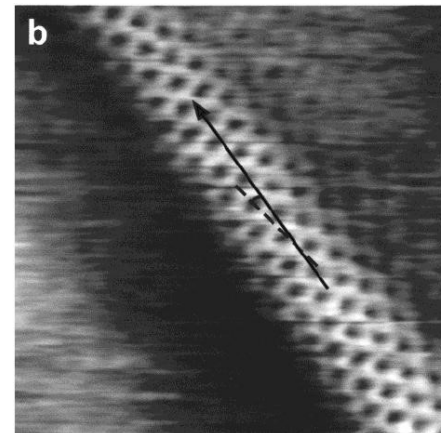


Carbon Nanotubes

CNT: Rolling-up a graphene sheet to form a tube



Schematic
of a CNT



STM image
of CNT

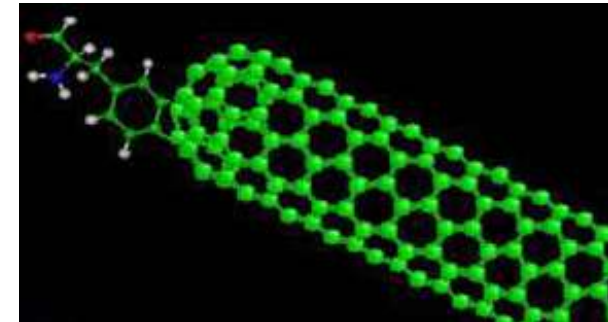


Carbon Nanotubes Property: Mechanical

- Carbon-carbon bonds are one of the strongest bond in nature
- Carbon nanotube is composed of perfect arrangement of these bonds
- Extremely high Young's modulus

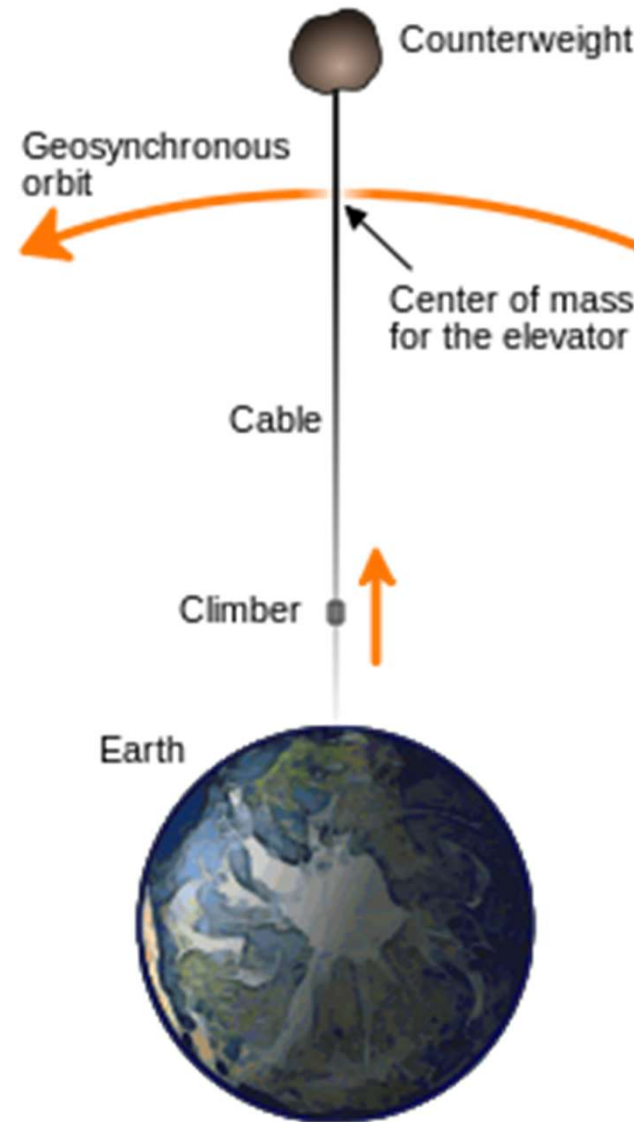
Material	Young's modulus (GPa)
Steel	190-210
SWNT	1,000+
Diamond	1,050-1,200

- Electrical conductivity six orders of magnitude higher than copper
- Can be metallic or semiconducting depending on chirality
 - 'tunable' bandgap
 - electronic properties can be tailored through application of external magnetic field, application of mechanical deformation...
- Very high current carrying capacity
- Excellent field emitter; high aspect ratio and small tip radius of curvature are ideal for field emission



Space Elevator

CNT enables the fabrication of space elevator due to its high Young's modulus, light weight and flexibility.

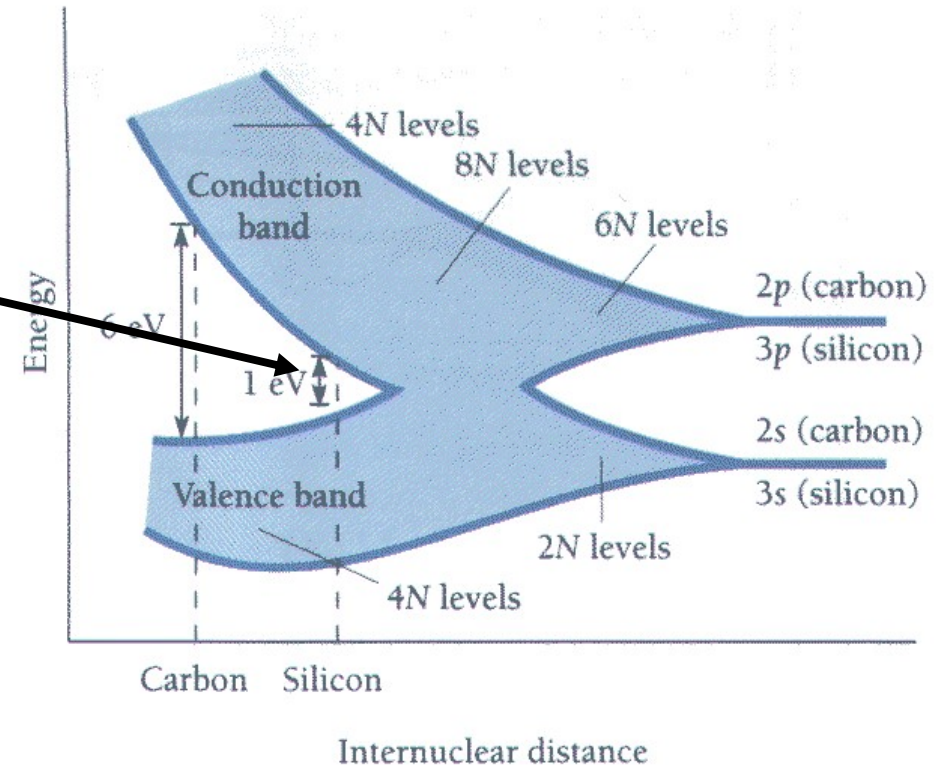


Silicon: Band Hybridization

Silicon has a similar band structure.
The forbidden gap is about 1 eV.



The probability of a single electron being excited across the gap is small, proportional to $\exp(-E_{\text{gap}}/k_B T)$.

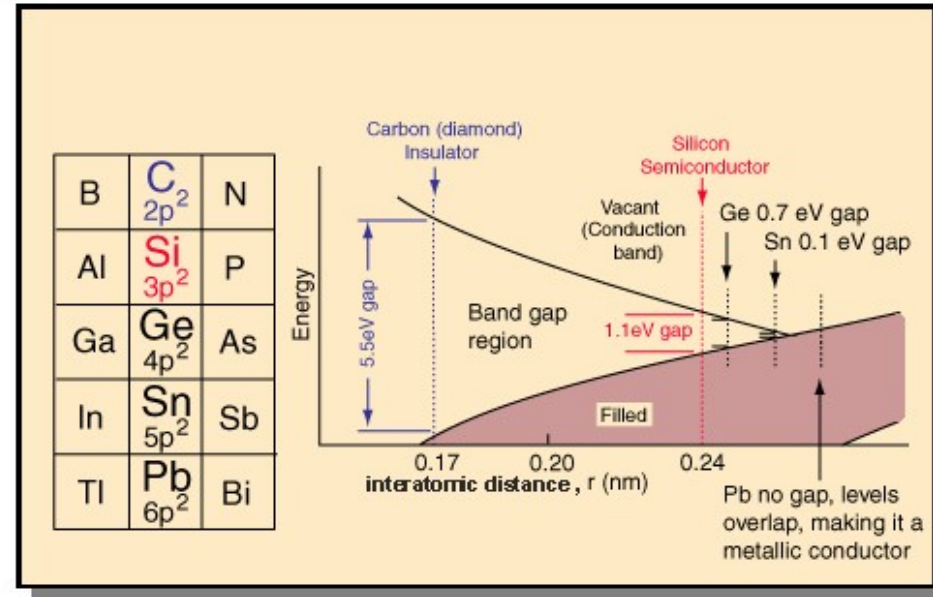
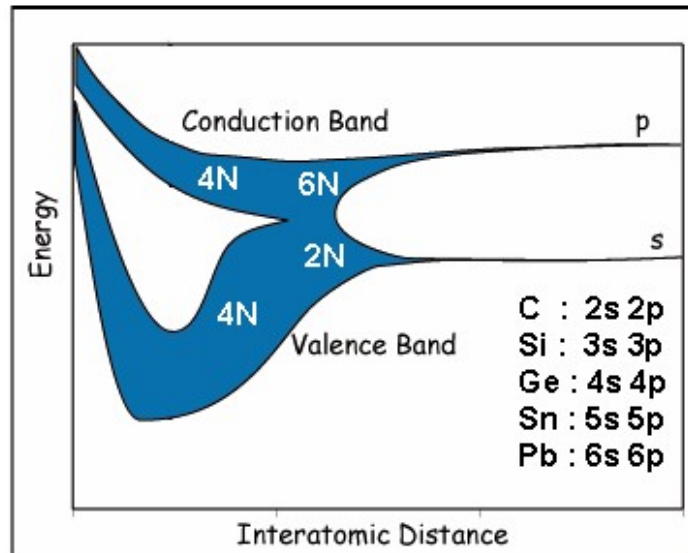


However, there are enough 3s+3p electrons in silicon that some of them might make it into the conduction band. We need to consider such a special case.



Energy Band of C, Si, Ge

C (diamond), Si, and Ge have similar band structures



Z larger $\rightarrow$ interatomic distance larger $\rightarrow$ bandgap smaller

Bandgap (E_g)

- C (diamond): ~ 6.0 eV
- Si: ~ 1.1 eV
- Ge: ~ 0.7 eV

